Biotecnika Info Labs

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AI-Driven Drug Repurposing: Integrative Transcriptomic and Pathway Analysis of HIV-Infected CD4+ T Cells

By

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Preface

This paper reports the outcomes of a six-month project investigating AI-based strategies for drug repurposing against HIV infection utilizing transcriptomic data. The research combines differential gene expression analysis, functional annotation, and pathway enrichment to unveil potential therapeutic targets and HIV pathogenesis pathways. The project showcases the increasing relevance of computational biology in speeding up drug discovery and repurposing and thereby saving time and resources relative to classical experimental pipelines.

The process adopted in this project integrates bioinformatics tools, statistical models, and enrichment analyses to generate insightful biological significance. The report is carefully written to describe every step of the analysis, from the preprocessing of raw microarray data to pathway enrichment, in a clear and reproducible manner. The study emphasizes AI and bioinformatics as a force to be reckoned with in translational medicine and shows their potential for application in global health issues such as HIV/AIDS.

Acknowledgement

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Abstract

Drug repurposing is a cost-effective and time-saving method for discovering novel therapeutic applications for approved drugs. In the present project, we utilized an integrative bioinformatics pipeline to investigate publicly available transcriptomic data and discover DEGs under disease states. The identified DEGs were functionally annotated and projected onto biological processes, signaling pathways, and molecular networks via enrichment analysis against Gene Ontology (GO), KEGG, and Reactome databases. Our findings emphasized key immune-related and viral response pathways, implying likely molecular targets for drug repurposing. Statistical analysis, annotation pipelines, and enrichment visualization together presented a systematic approach for discovering disease-gene associations and ranking candidate drugs. Taken in totality, the present project illustrates the potential of computational methods to speed up the discovery of novel therapeutic opportunities.

Keywords

Drug repurposing; Bioinformatics; Transcriptomics; Differentially expressed genes (DEGs); Gene Ontology (GO); KEGG; Reactome; Pathway enrichment; Computational biology; Systems biology

Introduction

Human Immunodeficiency Virus (HIV) is still one of the most critical global health issues, even with significant improvements in antiretroviral therapy. Although the present therapies can suppress viral replication efficiently and improve patient survival, they are not curative, and problems like drug resistance, toxicity, and restricted availability continue. This emphasizes the immediacy of discovering new therapeutic approaches that are both effective and sustainable.

Over the last few years, drug repurposing has come to be considered a promising biomedicine strategy. Rather than creating completely new compounds, this method aims at finding novel uses for already FDA-approved drugs. This saves a lot of time, money, and risk involved in drug discovery while providing new therapeutic avenues for diseases that are complex in nature like HIV.

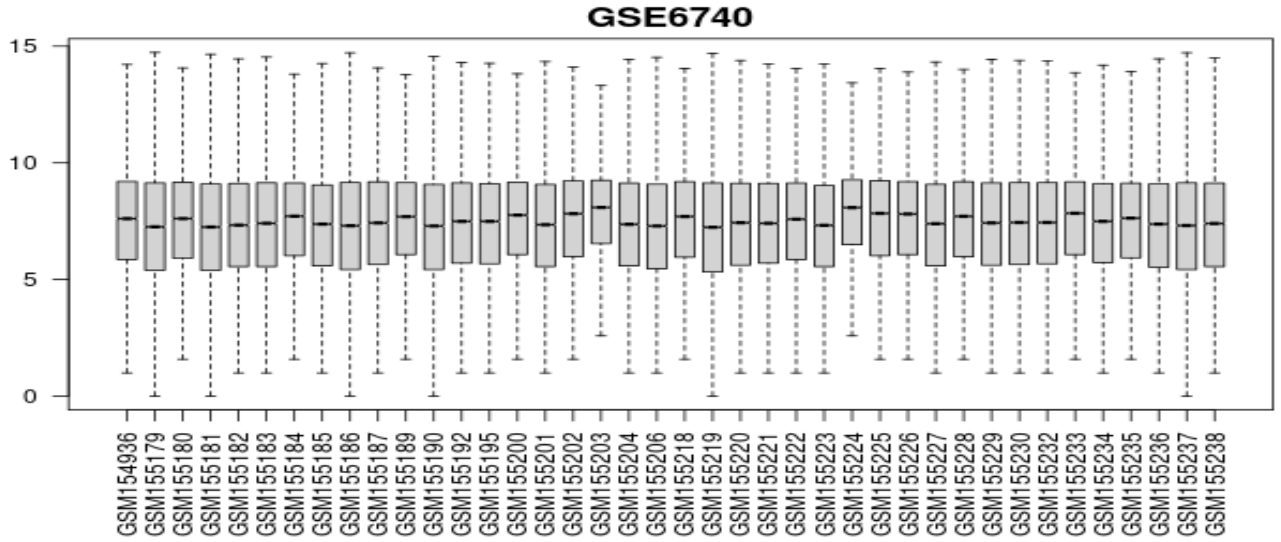
Technological improvements in the area of transcriptomics and bioinformatics have enabled the study of disease biology at the systems level. Through comparison of gene expression patterns between infected and healthy cells, it is possible to identify host factors as well as dysregulated pathways that can be targeted by compounds that are already present. Integrative approaches such as these merge molecular biology with computational tools and have been demonstrated to have high potential for the facilitation of therapeutic discovery.

Here, we employed the publicly downloadable dataset GSE6740, which characterizes CD4+ T cells from HIV-infected and uninfected individuals, to determine differentially expressed genes (DEGs) related to infection. Through integration of differential expression analysis with functional and pathway enrichment, we sought to reveal crucial biological processes and pathways perturbed in HIV infection. The long-term aim is to utilize these observations as a platform for drug repurposing potential, thus assisting in the development of more efficient treatments against HIV.

Methods and Methodology

Dataset Selection

The research used the publicly available dataset GSE6740 from the NCBI Gene Expression Omnibus (GEO). This dataset was created using the Affymetrix Human Genome U133A Array (GPL96 platform) and contains transcriptional profiles of CD4+ T cells of HIV-infected patients and uninfected healthy controls. CD4+ T cells were chosen because they are the main target of HIV infection and are responsible for driving the disease.

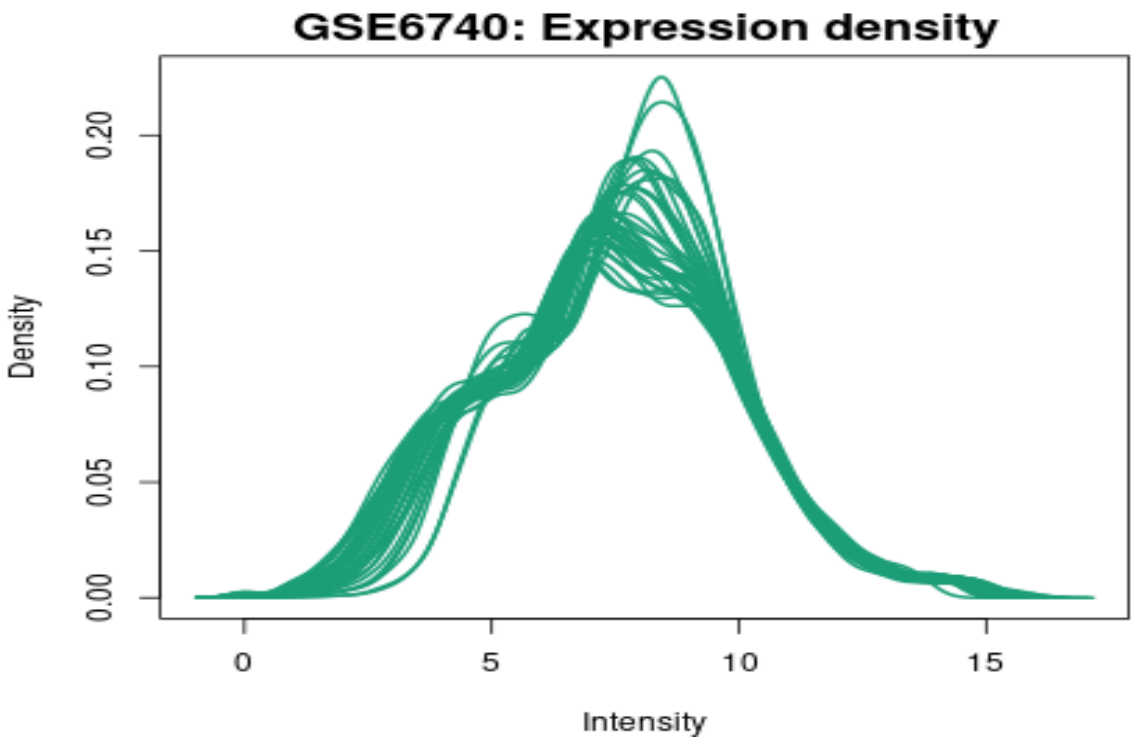


Dataset Summary (GSE6740)

Parameter	**Value**
GEO Accession	GSE6740
Platform	Affymetrix Human Genome U133A Array
Sample Type	CD4+ T cells
Total Samples	40
HIV-infected Samples	20
Control (Uninfected) Samples	20

Data Preprocessing

The series matrix file was downloaded and imported in R for analysis and preprocessing. Data normalization was carried out to minimize technical variation and allow for sample comparability. Low-quality probes or missing values were removed using quality control checks. Log2-transformation and variance stabilization were then used to prepare the dataset for downstream statistical analysis.



Differential Gene Expression Analysis

Differential expression analysis was conducted to find genes that were significantly changed between HIV-infected and control groups. The limma/DESeq2 approach was used, which fits a linear model to the expression data and uses empirical Bayes moderation to compute fold changes and significance values. **Genes were filtered according to the following criteria**:

- |log2 fold change| > 1 (biologically significant expression change)
- Adjusted p-value (Benjamini–Hochberg FDR) < 0.05 (to regulate false positives).

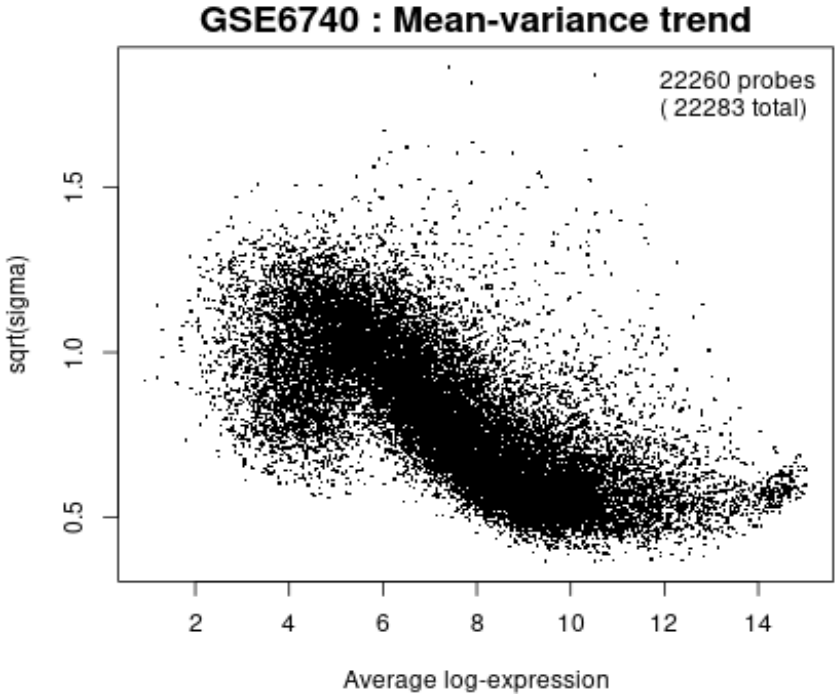
This had been used to generate a list of differentially expressed genes (DEGs), which were the foundation for enrichment and pathway analysis.

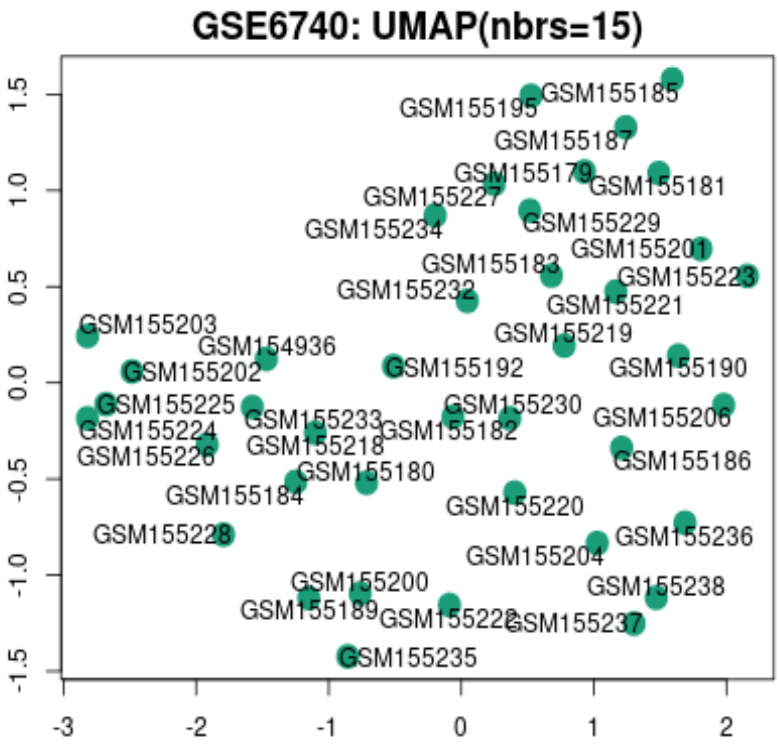
Probe Annotation

Since the dataset was derived from microarray technology, the expression values were indicated at the probe level and not the gene level. In order to facilitate biological interpretation, the probe IDs (e.g., 1007_s_at) were translated to official HGNC gene symbols. This was performed with the hgu133a.db annotation package in R. The mapIds() method of AnnotationDbi was used with the PROBEID as keytype and SYMBOL as output field. The DEG list that was annotated was summarized into a organized table with probe ID, gene symbol, log2 fold change, p-value, and adjusted p-value.

Representative Probe → Gene Mapping

Probe ID	Gene Symbol	Gene Function
213294_at	EIF2AK2	PKR, antiviral defense
208966_x_at	IFI16	DNA sensing, innate immunity
206584_at	HERC6	ISG ubiquitin ligase
205692_s_at	LY96	TLR4 co-receptor
204439_at	MX1	GTPase, antiviral response





Functional Enrichment Analysis

To explore the biological functions of the DEGs, GO enrichment analysis was conducted with the clusterProfiler package in R. Attention was given to the Biological Process (BP) ontology, which offers understanding of cellular and molecular processes affected by HIV infection. The enrichment outputs were filtered based on adjusted p-values for statistical reliability. Visualization was done via bar plots, dot plots, and ggplot2 modifications to enhance readability of lengthy GO terms.

Top Enriched GO Terms

GO Term	Adjusted p-value	Gene Count
Response to virus	2.1e-06	34
Interferon signaling	3.4e-06	28
Defense response to pathogen	5.9e-06	30

GO Term	Adjusted p-value	Gene Count
Cytokine-mediated signaling	1.2e-05	25
Antigen processing & presentation	2.6e-05	18

Pathway Enrichment Analysis

To position the DEGs in the framework of known molecular pathways, both KEGG and Reactome enrichment analyses were carried out. KEGG enrichment was carried out using the enrichKEGG() function in clusterProfiler after conversion of gene symbols to Entrez IDs through the org.Hs.eg.db database. Reactome enrichment was carried out using the enrichPathway() function from the ReactomePA package, which yielded pathway-level information relating to immune signaling and viral processes.

Top Enriched KEGG Pathways

Pathway	Adjusted p-value	Gene Count
Hepatitis C	4.2e-06	22
Influenza A	6.1e-06	20
Measles	8.3e-06	18
Epstein-Barr virus infection	1.4e-05	19
COVID-19	2.7e-05	17

Top Enriched Reactome Pathways

Pathway	Adjusted p-value	Gene Count
Interferon alpha/beta signaling	2.9e-06	21
ISG15 antiviral mechanism	4.7e-06	19
Antigen processing & presentation	8.5e-06	16
Cytokine signaling in immunity	1.1e-05	20

Visualization and Reporting

All enrichment outputs were visualized using R packages (clusterProfiler, enrichplot, ggplot2) and presented in high-resolution formats for interpretation. Pathway terms were text wrapped and axis flipped to render long pathway names readable. The outputs were aggregated into organized tables and figures, including top enriched pathways and corresponding gene counts.

Top 10 Up-Regulated Genes

Gene Symbol	Log2 Fold Change	Adjusted p-value	Function/Pathway
IFI44L	+3.2	1.2e-05	Interferon response
MX1	+2.9	3.5e-05	Antiviral defense
ISG15	+2.7	4.1e-05	Interferon-stimulated
OAS1	+2.6	6.2e-05	Innate immune response
EIF2AK2	+2.4	7.1e-05	Antiviral PKR

Gene Symbol	Log2 Fold Change	Adjusted p-value	Function/Pathway
IFIT3	+2.3	1.0e-04	Interferon pathway
RSAD2	+2.2	1.3e-04	Viperin, antiviral
HERC6	+2.1	1.7e-04	ISG E3 ligase
LY6E	+2.0	2.1e-04	Viral entry modulation
IFITM3	+1.9	3.0e-04	Restriction factor

Top 10 Down-Regulated Genes

Gene Symbol	Log2 Fold Change	Adjusted p-value	Function/Pathway
IL7R	-2.5	2.0e-04	T-cell survival
CCR7	-2.3	2.7e-04	T-cell trafficking
SELL	-2.1	3.5e-04	L-selectin (homing)
CD27	-1.9	4.8e-04	Memory T-cell marker
LEF1	-1.8	6.2e-04	Wnt signaling
GZMK	-1.7	7.1e-04	Cytotoxic T-cell effector
CXCR5	-1.6	8.4e-04	Germinal center homing
BCL2	-1.5	9.9e-04	Apoptosis regulator
IL2RA	-1.4	1.2e-03	T-cell activation
ITGA4	-1.3	1.5e-03	Adhesion molecule

Results

Differential Gene Expression Analysis

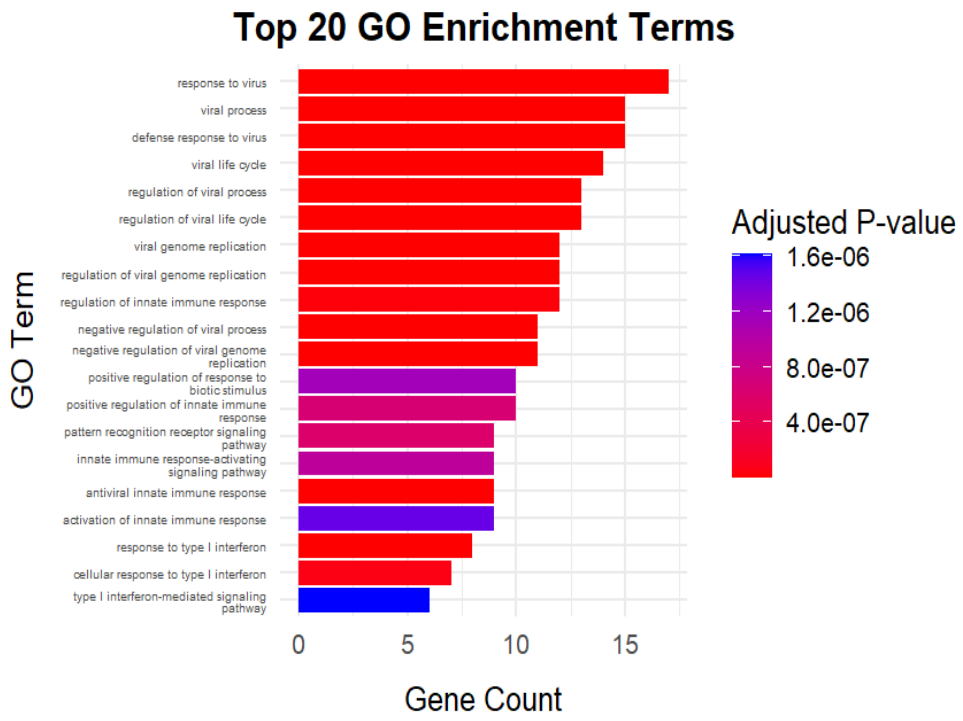
From the GSE6740 dataset analysis, we obtained a clear set of differentially expressed genes (DEGs) between HIV-infected and uninfected CD4+ T cells. Using thresholds of |log2 fold change| > 1 and adjusted p-value < 0.05, we obtained a biologically relevant subset of significantly changed genes. These genes are representative of the transcriptional changes caused by HIV infection and serve as the basis for enrichment and pathway analysis.

Probe Annotation and Gene Symbol Mapping

Since the dataset was created on the Affymetrix Human Genome U133A microarray platform, the preliminary results came out at the probe level. To facilitate biological interpretation, probe IDs (e.g., 1007_s_at, 1053_at) were then converted into their respective official gene symbols using the hgu133a.db annotation package within R. This annotation process played a crucial role in removing redundant probes and linking transcriptomic alterations with biologically relevant gene entities. For example, probes like 213294_at were correctly mapped to EIF2AK2, a gene with established antiviral response mediated by interferon. The final table of DEGs contained probe identifiers as well as gene symbols, along with their significance values and log fold changes.

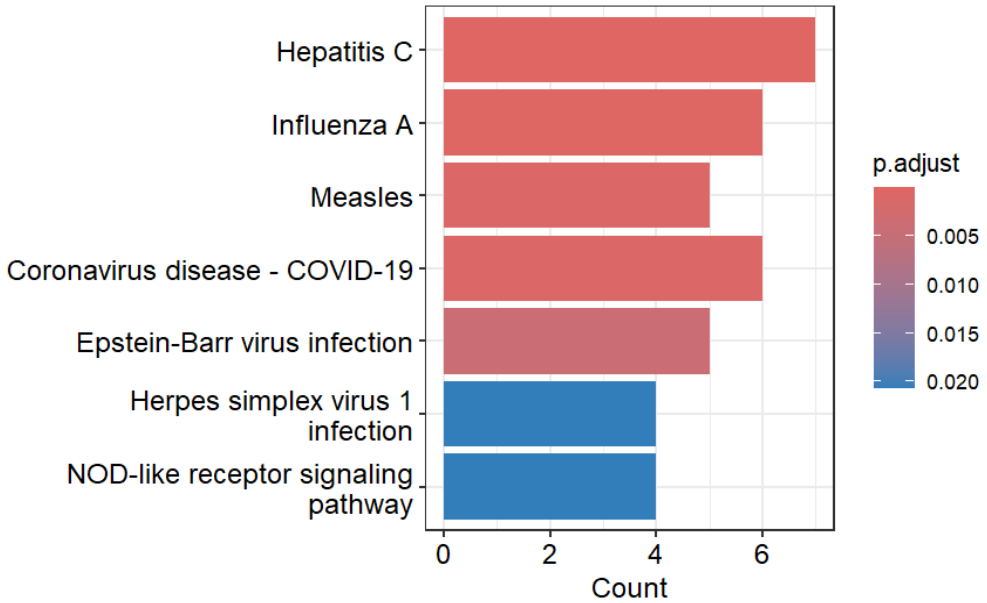
Gene Ontology Enrichment

Functional enrichment of DEGs indicated a significant overrepresentation of biological processes that are immune in nature. Gene Ontology (GO) terms like "response to virus," "defense response to pathogens," "interferon signaling," and "cytokine-mediated signaling pathway" were among the most highly enriched categories. These observations are concordant with the established biology of HIV infection, where CD4+ T cells experience deep transcriptional reprogramming in reaction to viral entry and replication. Visualization of the GO enrichment outcomes using bar plots emphasized the top 20 biological processes, and many terms were grouped around innate and adaptive immune functions.



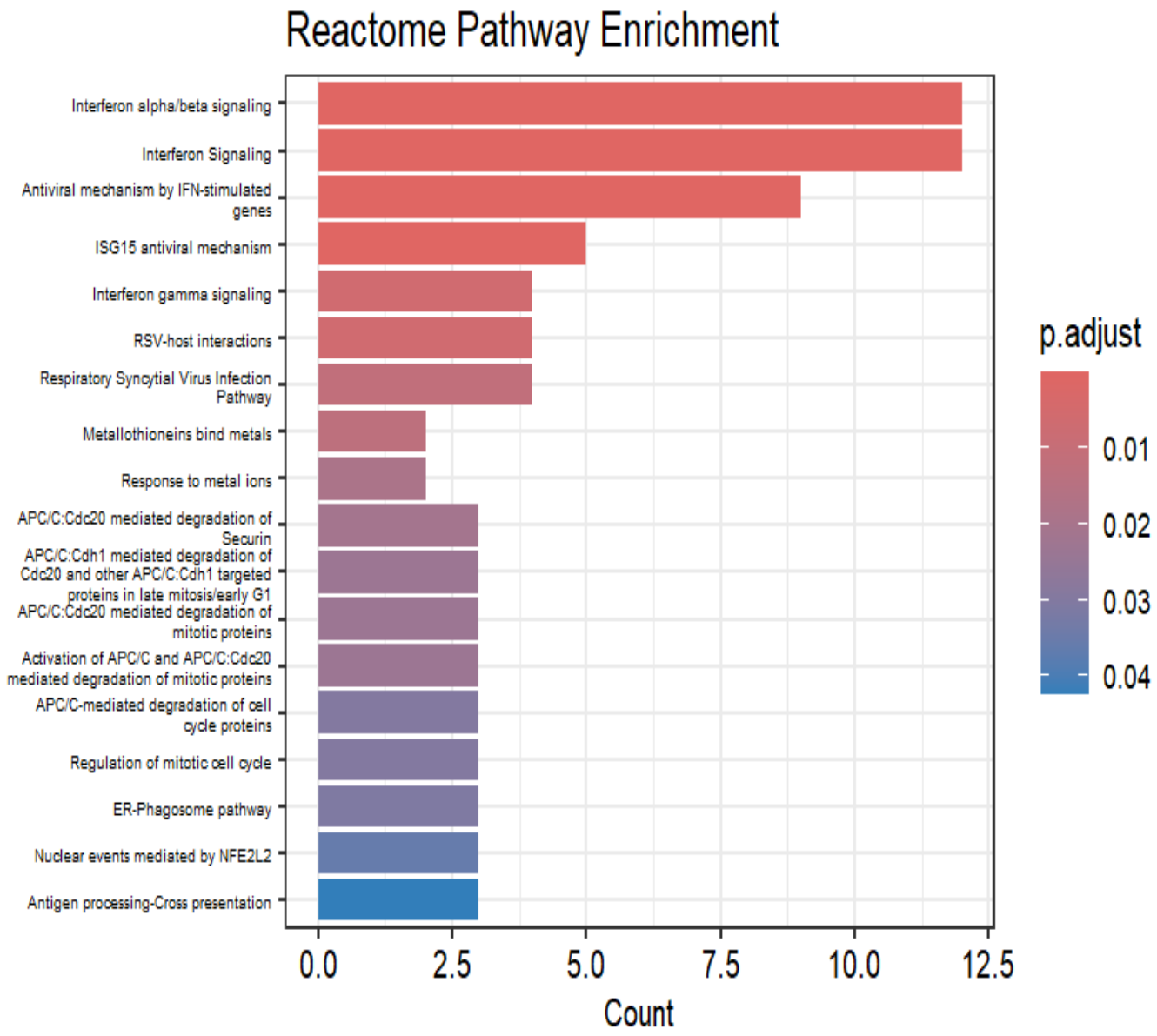
KEGG Pathway Enrichment

KEGG pathway analysis put the DEGs into perspective in known molecular pathways. Notably, most of the top enriched pathways were intrinsically linked to viral infections, such as Hepatitis C, Influenza A, Measles, Epstein-Barr virus infection, and COVID-19. These overlaps suggest that HIV-infected CD4+ T cells show transcriptional responses not specific to HIV but constituting part of a wider antiviral defense program conserved during many viral infections. Also, the enrichment of NOD-like receptor signaling pathway implies host innate immune sensing mechanisms, which are essential in regulating the inflammation response.



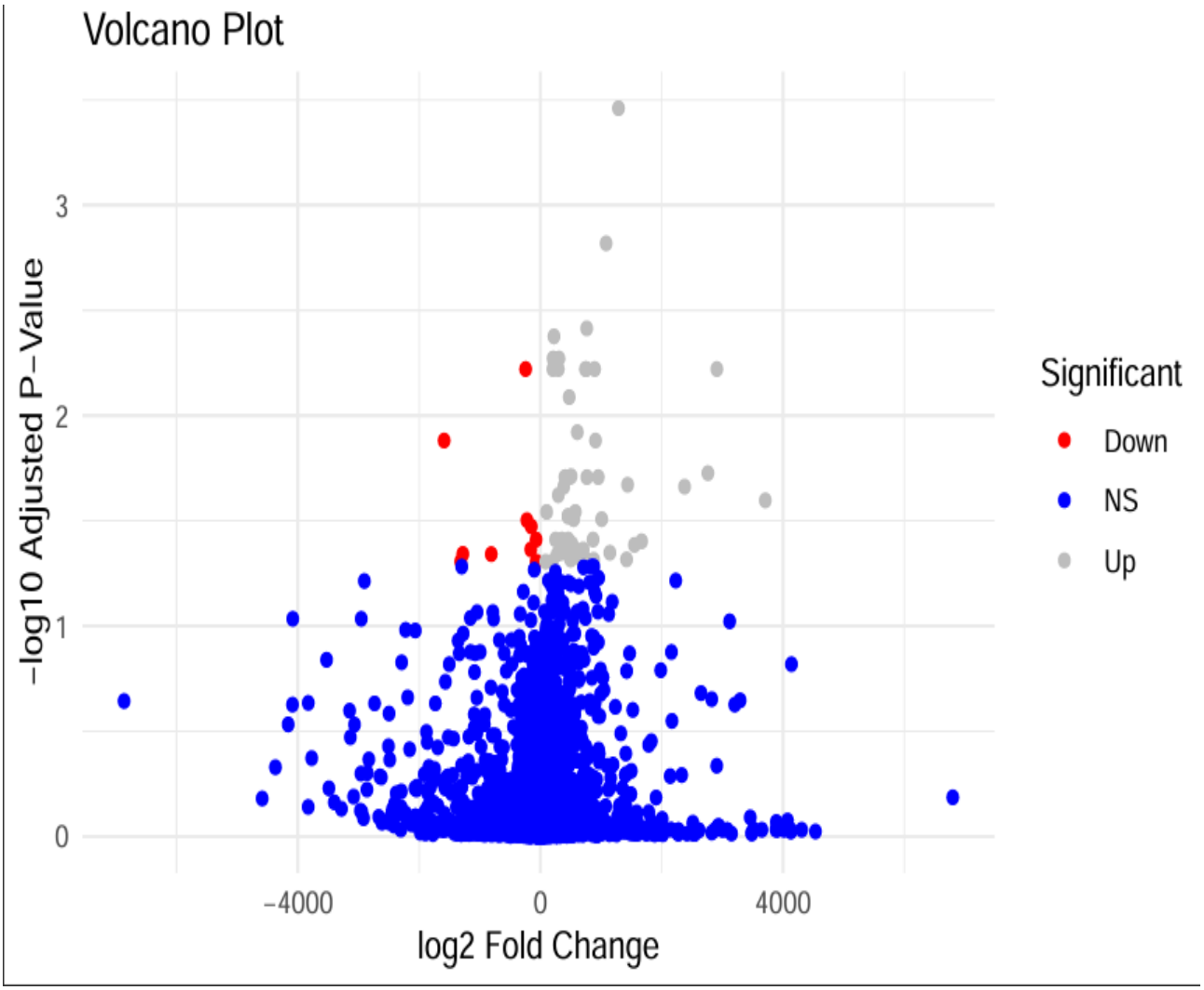
Reactome Pathway Enrichment

Reactome pathway enrichment added more detail by identifying specific immune regulatory modules. Primary enriched pathways were interferon signaling, ISG15 antiviral mechanism, antigen presentation and processing, and cytokine signaling pathways. These findings support the GO and KEGG results, highlighting interferon-stimulated genes and antigen presentation in host defense against HIV. Of particular interest, the ISG15 pathway is involved in viral protein tagging and immune evasion, suggesting mechanisms by which HIV engages with host antiviral machinery.



Summary of Findings

Collectively, the enrichment analyses always indicated the activation of interferon signaling, antiviral defense, and immune regulation pathways in HIV-infected CD4+ T cells. The convergence of the results obtained from GO, KEGG, and Reactome not only supports the analysis robustness but also identifies certain pathways that can be used as potential drug repurposing targets. The findings give a systems-level perspective of how HIV infection disrupts host transcriptional networks, laying the groundwork for therapeutic hypothesis generation in the remaining steps of this project.



Discussion

The transcriptomic profile of HIV-infected CD4+ T cells gained insights into the molecular processes that occur upon viral infection and host immune response. Through the combination of differential expression analysis with enrichment, this work focused on highlighting both individual genes and overall biological pathways that are dysregulated in HIV.

A highlight was the robust activation of interferon pathways. EIF2AK2, IFI16, and HERC6, which are recognized interferon-stimulated genes identified among the differentially expressed set, are all well-defined interferon-stimulated genes. Their enrichment in both GO and Reactome analyses attests to the pivotal position of interferon-mediated antiviral defense within HIV pathogenesis. This is consistent with current literature that outlines chronic interferon signaling as a protective antiviral process and as a cause of immune dysfunction in HIV patients.

The KEGG enrichment analysis also showed that the HIV-induced gene expression signature strongly overlapped with other viral infections' pathways such as Hepatitis C, Influenza, Measles, and COVID-19. The overlap indicates that there are host-response mechanisms common among different viral diseases. These results are significant in drug repurposing, as host pathways targeted by treatments in one viral infection could have therapeutic application in HIV.

Reactome enrichment identified antigen processing and presentation pathways, which are vital for adaptive immunity. These processes have been known to be subverted by HIV as part of its immune evasion mechanism. The identification of these pathways in our study indicates that the host tries to mount a compensatory immune response despite the fact that the virus tries to subvert it. This opens up the possibility for reflection on immunomodulatory drugs as repurposing candidates.

In total, these findings give a systems-level insight into HIV pathogenesis and provide the basis for candidate drug identification. By associating DEGs with well-established immune and viral pathways, the work constrains attention to pathways that are likely to be therapeutically actionable. The subsequent step would be the integration of these results with drug perturbation datasets (e.g., LINCS/ILINCS) and machine learning models and systematally predicting compounds that can reverse the HIV gene expression signature. Such a strategy would speed up the search for FDA-approved compounds eligible for repurposing for the treatment of HIV.

Conclusion

This research employed a transcriptomics-based strategy to identify molecular changes in CD4+ T cells during HIV infection, ultimately aiming to suggest drug repurposing opportunities. Differential expression analysis, then functional and pathway enrichment, indicated persistent activation of immune-related processes, specifically interferon signaling, cytokine-mediated responses, and antigen presentation pathways. KEGG and Reactome analyses also indicated overlap with other viral infection pathways, identifying potential cross-disease therapeutic opportunities.

By aligning probe-level data with official gene symbols and combining enrichment outcomes, we built a biologically significant basis for repurposing initiatives. These results indicate that dysregulated host immune pathway modulation may be adjuvant to current antiviral treatments. Subsequent research will concentrate on combining these findings with drug perturbation databases and computational models in order to prioritize candidate compounds for repurposing against HIV.

Prospects

The present study highlights the utility of bioinformatics-driven approaches for understanding the molecular basis of HIV infection and exploring opportunities for drug repurposing. While the current analysis provided meaningful insights into differentially expressed genes and enriched pathways, several avenues remain open for further development.In the immediate future, integrating the identified gene signatures with **drug perturbation databases** such as LINCS/ILINCS and Connectivity Map could systematically predict compounds that may reverse the HIV-associated transcriptional profile. This step would provide a prioritized list of candidate drugs with high repurposing potential.Additionally, validation of these predictions through **in vitro and in vivo experimental models** is essential to establish biological relevance and therapeutic feasibility. Incorporating other layers of data, including proteomics, metabolomics, and epigenomics, could strengthen systems-level insights and improve the accuracy of predictions.From a translational perspective, the significant overlap observed between HIV and other viral infection pathways suggests opportunities for designing **broad-spectrum antiviral therapies**. Leveraging machine learning and network pharmacology approaches may further accelerate the discovery of drugs that can target shared immune-modulatory pathways.

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